

Connect and Protect

Mastering Networks and Network Security

An Executive Overview of the
Google Cybersecurity
Professional Certificate Module



The Global Standard in Cyber Training

A foundational blueprint designed by Google to transform beginners into highly capable network security professionals—all on a flexible, self-paced schedule.



Scale

506,918
Enrolled



Quality

4.8/5 Rating
(7,177 Reviews)



Impact

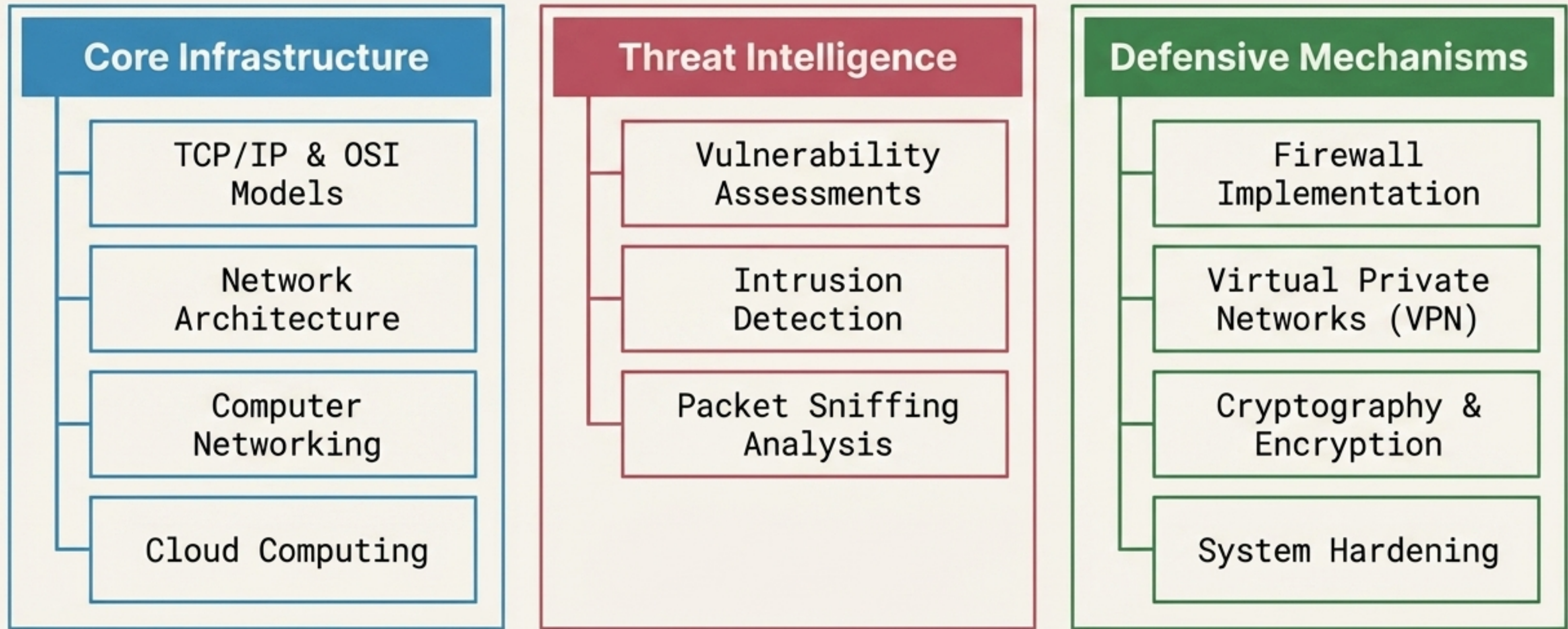
98%
Learner Satisfaction



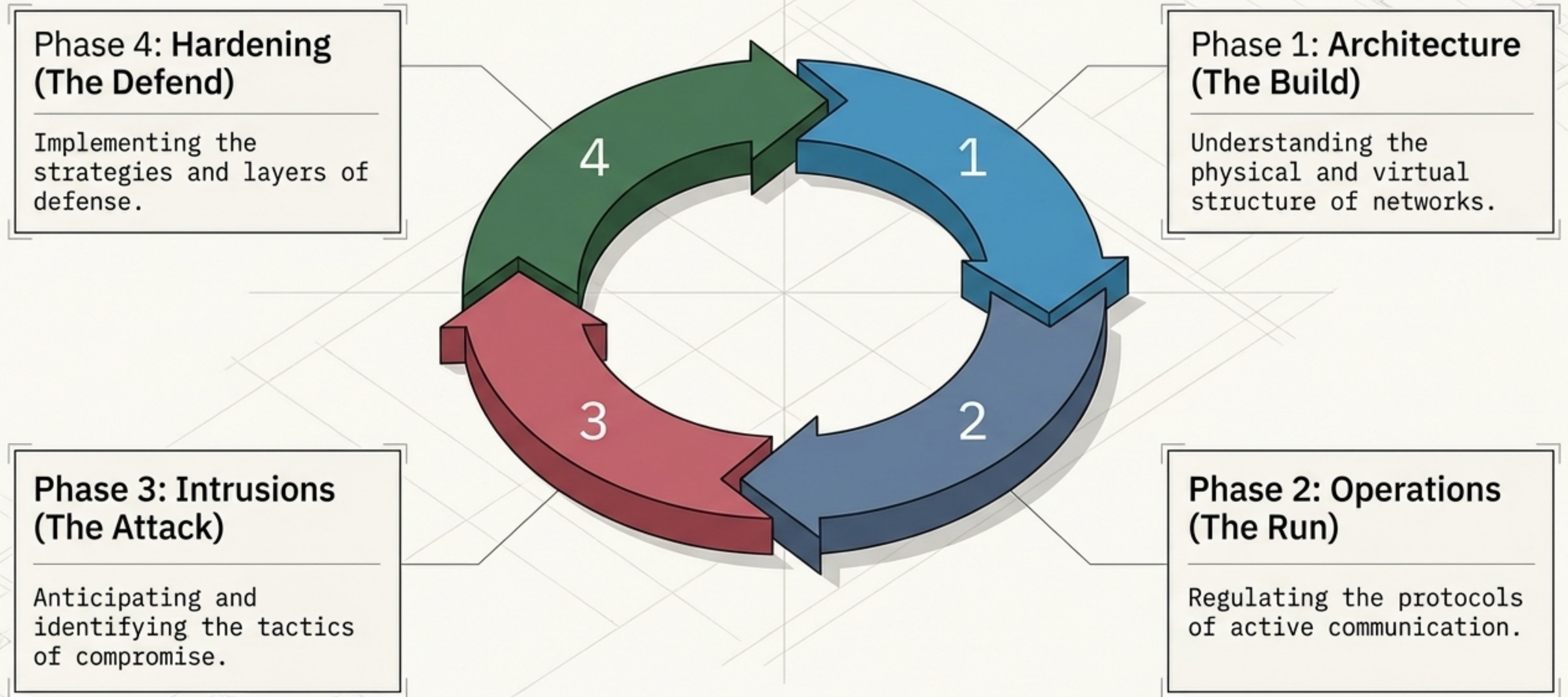
Commitment

1 Week Timeline
10 Hours Total Effort

The Cybersecurity Capability Matrix



The 4-Phase Learning Lifecycle



Phase 1: Network Architecture

The Build (3 Hours to Complete)

Core Focus: Understanding the fundamental mechanisms and structures that allow networks to communicate securely.

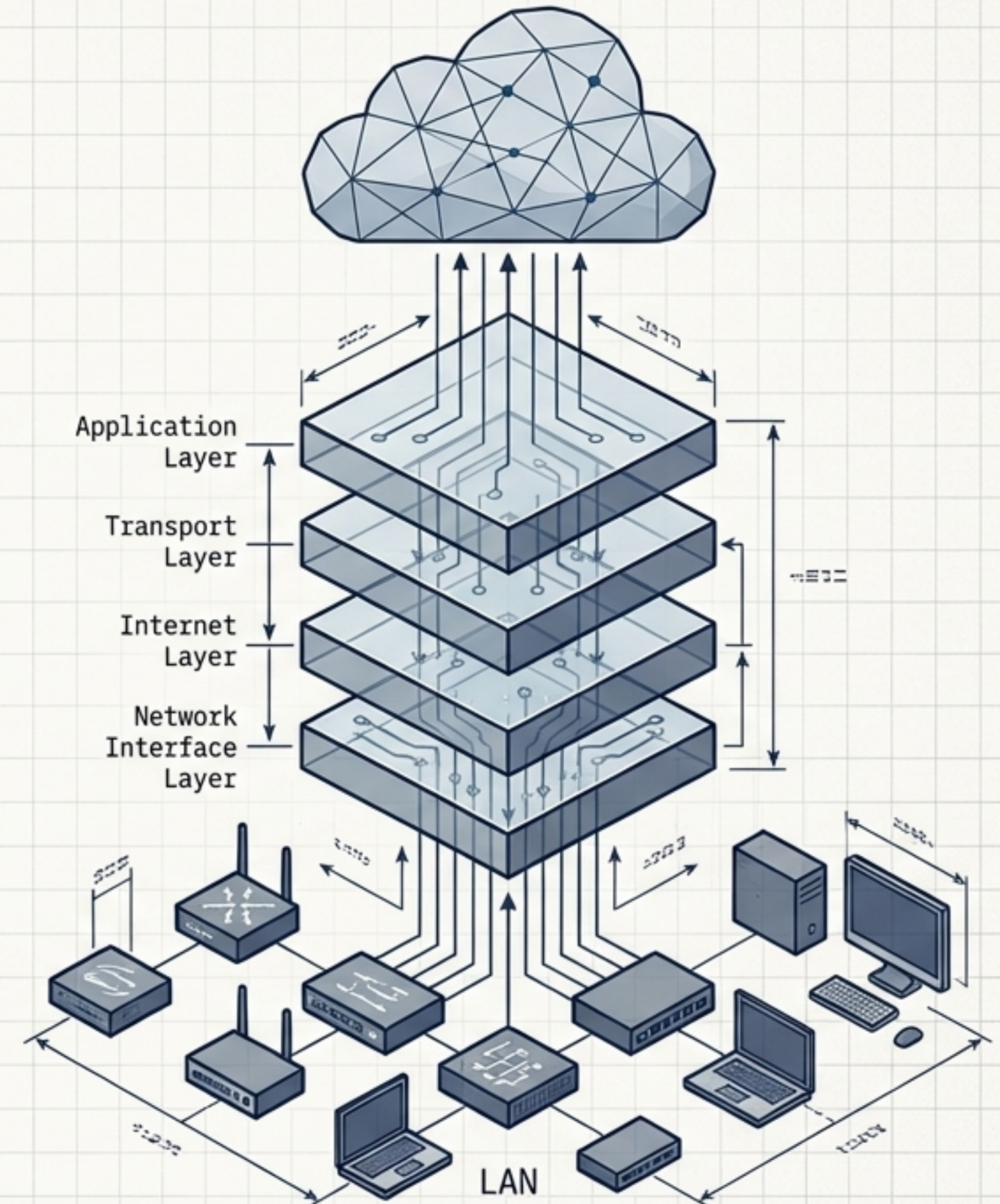
Key Concepts

The 4 layers of the TCP/IP model

The OSI model structure

Cloud computing vs. Software-Defined Networks

Hands-On Application:
Design a local area network



Phase 2: Network Operations

The Run (2 Hours to Complete)

Core Focus:

Regulating and protecting active communication protocols to ensure safe, reliable network operations.

Key Concepts

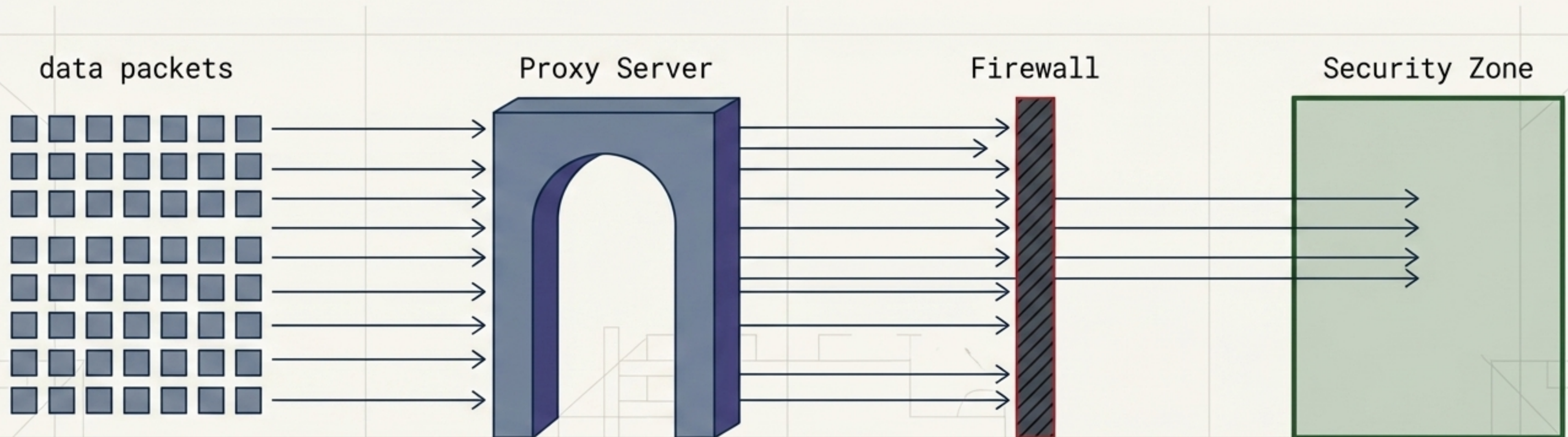
Subnetting and CIDR implementation

Wireless security protocol evolution

VPN protocols:
Wireguard and IPSec

Security Measures

Implementation of Firewalls, Proxy Servers, and isolated Security Zones.



The Infrastructure Duality: Local vs. Cloud



Local Network



Cloud Computing

Architecture	Physical hardware, tangible routers/switches.	Virtualized infrastructure, software-defined perimeters.
Primary Vulnerabilities	Physical access breaches and local loop vulnerabilities.	Shared tenancy risks and system misconfigurations.
Defensive Posture	Hardened edge firewalls and physical zoning.	Advanced Cryptography and Identity/Access Management (IAM).

Phase 3: Securing Against Intrusions

The Attack (3 Hours to Complete)

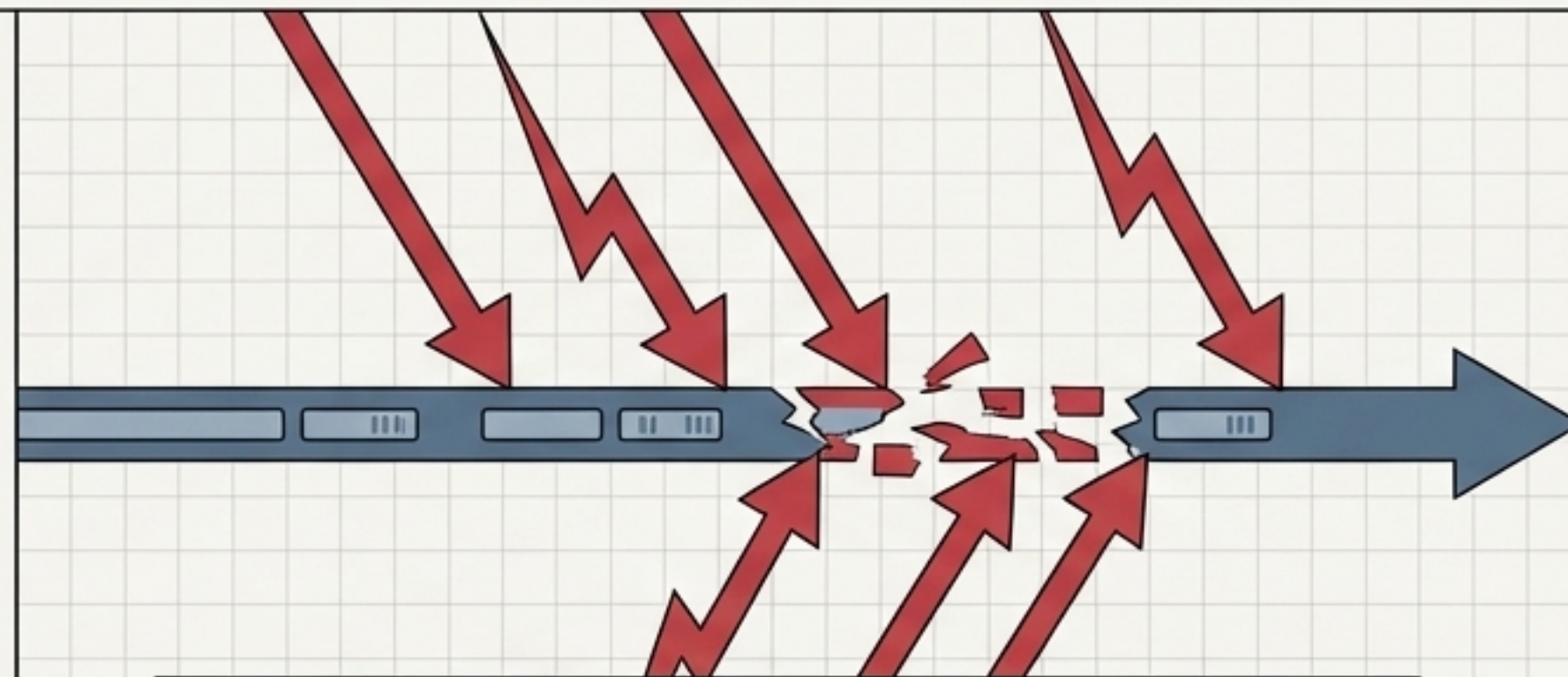
Anticipating malicious tactics and identifying vulnerabilities within network infrastructure.

Threat Vectors Analyzed

Denial of Service (DoS & DDoS)

Malicious Packet Sniffing

IP Spoofing & Network Interception



Offensive Playbook

```
14:32:10.551289 IP 192.168.1.100.5000 > 10.0.0.5.443: Flags [S],  
seq 23456789, win 65535, options [mss 1460, sackOK, TS val 123456  
ecr 0, nop, wscale 7], length 0  
14:32:10.553412 IP 18.0.0.5.443 > 192.168.1.100.5000: Flags [S.],  
seq 98765432, ack 23456790, win 65535, options [mss 1460, sackOK,  
TS val 654321 ecr 123456, nop, wscale 7], length 0  
14:32:10.555890 IP 192.168.1.100.5000 > 10.0.0.5.443: Flags [A],  
seq 1, ack 1, win 65535, length 0  
14:32:15.120981 IP 172.16.1.20.3000 > 10.0.0.5.22: Flags [S], seq  
11223344, win 65535, options [mss 1460], length 0 -- SUSPICIOUS  
ACTIVITY DETECTED
```

Hands-On Application: Reading tcpdump logs and analyzing simulated network layer communications to identify breaches.

Phase 4: Security Hardening

The Defend (3 Hours to Complete)

Strengthening systems through preliminary measures to neutralize future breaches.

OS Hardening

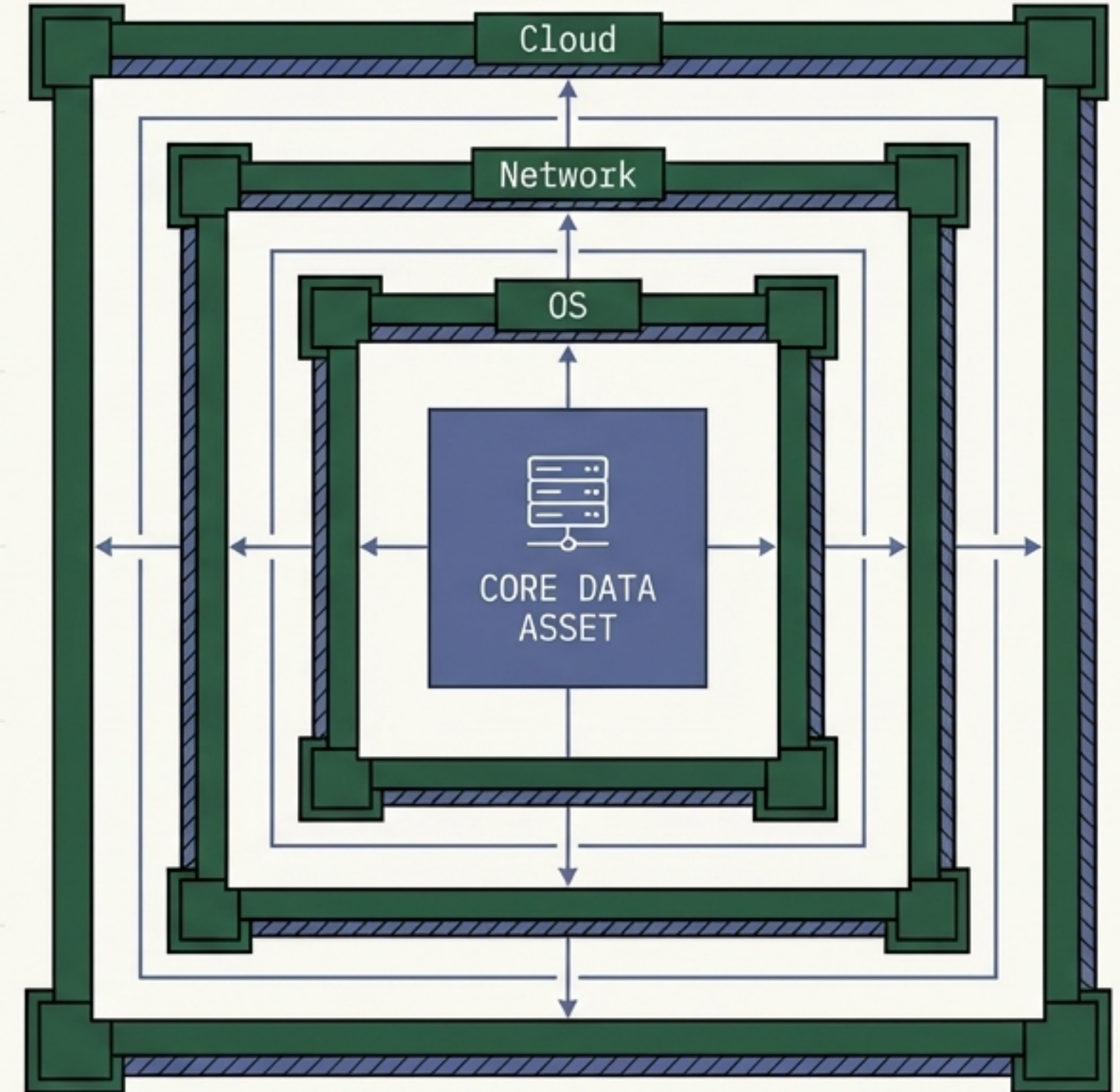
Techniques to mitigate brute force attacks at the operating system level.

Network Hardening

Preliminary infrastructure measures to secure active applications.

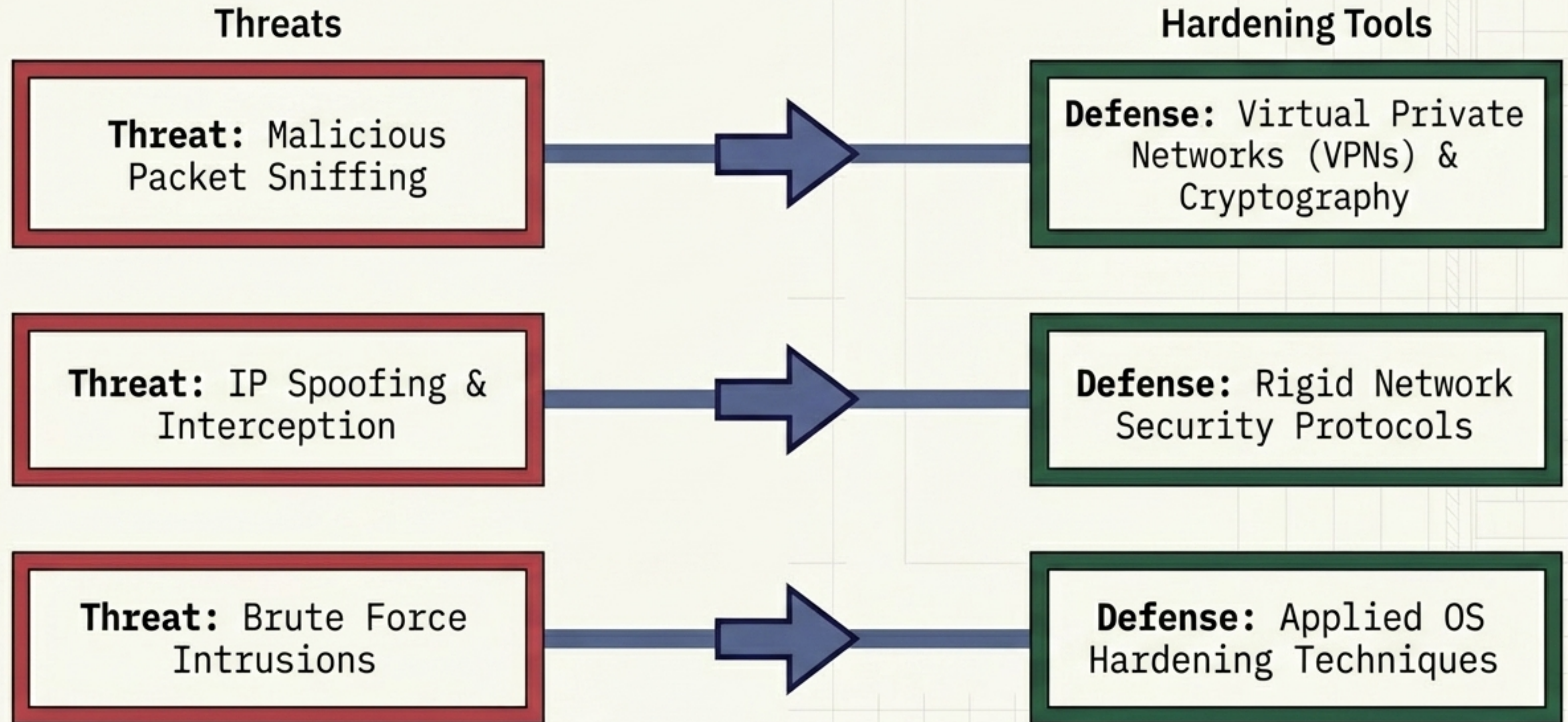
Cloud Hardening

Applying cryptography to secure decentralized cloud infrastructure.



FORTRESS SHIELDING ARCHITECTURE

Synthesis: Threat & Threat & Defense Alignment



Portfolio Building & Real-World Application

Learners complete 19 distinct assignments, generating verifiable artifacts rather than just theoretical knowledge.

THE SHOWPIECE	
Use the NIST Cybersecurity Framework to respond to a security incident.	

THREAT SIMULATION	
Analyzing logs from a real-life DDoS attack.	

SYSTEM DEFENSE	
Active application of OS hardening techniques to a vulnerable operating system.	

The ROI: Individual Mastery & Enterprise Scale

For the Learner

- Earn a prestigious Google credential.
- Master in-demand tools like firewalls and tcpdump.
- Add verified skills directly to your LinkedIn profile.

For the Enterprise

- Upskill teams efficiently utilizing Coursera for Business—trusted by 3,400+ global companies to build the digital workforce of tomorrow.



**Access the best of Google's training
to build the workforce of tomorrow.**